Lakshmi Pakala

AI / Machine Learning Engineer

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LinkedIn — GitHub

Professional Summary

Generative AI and NLP-focused engineering graduate with hands-on experience in Python, machine learning, and LLM-powered applications, including RAG pipelines, vector embeddings, and Hugging Face models. Experienced in audio feature extraction, model evaluation, and building interactive AI applications with Streamlit.

Technical Skills

Programming: Python, SQL
Machine Learning: Scikit-learn, supervised learning, model evaluation, feature extraction, preprocessing
Deep Learning: CNN, TensorFlow, Keras
Generative AI: LangChain, Groq API, Ollama, RAG, vector embeddings, prompt engineering, LLM apps
NLP: NLTK, text preprocessing, Hugging Face
Development: Streamlit, Flask, React.js, HTML, CSS, JavaScript
Deployment: Streamlit Community Cloud, Git, GitHub
Tools: Jupyter Notebook, Google Colab, VS Code

Experience

Prompt Engineering & Generative AI Intern *May 2025 – July 2025*
Virtual Internship, Tirupati

- Practiced prompt engineering techniques for Generative AI applications and LLM-based workflows.
- Explored structured prompting and evaluated model responses for relevance and usefulness.

Digital Media Intern – AI Frame Artist *Feb 2026 – Jun 2026*
Sri City, Andhra Pradesh

- Supported AI-assisted visual content production and reviewed approximately 50–100+ generated frames per review cycle.
- Evaluated visual consistency and quality to help maintain content requirements.

Projects

RAG Document Q&A (GitHub) (Live Demo)

- Built an app that answers questions from uploaded PDFs and text files using only their content, with LangChain, Hugging Face embeddings, FAISS retrieval and Groq LLMs.

MathsGPT – LLM-Powered Math & Coding Assistant (GitHub) (Live Demo)

- Built an assistant that solves math word problems and coding questions step by step, using a calculator tool for exact arithmetic and Wikipedia search for facts (LangChain, Groq, Streamlit).
- Returns explained solutions with working code and time-complexity analysis for algorithm questions, such as the sliding-window approach for the longest substring.

Neonatal Cry Analysis (GitHub) (Live Demo)

- Built a classifier that identifies why an infant is crying (hungry, tired, belly pain, burping, discomfort) from audio, using MFCC features and a CNN trained on 1,858 recordings, with live WAV prediction in Streamlit.
- Compared the CNN against SVM and Decision Tree baselines on a held-out test set.

Education

B.Tech – Artificial Intelligence and Machine Learning *2022 – 2026*
Geethanjali Institute of Science and Technology, Nellore CGPA: 8.29/10

Intermediate – MPC *2020 – 2022*
Sri Chaitanya Junior College, Nellore CGPA: 9.09/10

Certifications

- Python Programming – CodeChef
- SQL – LinkedIn Learning
- Web Technologies Using Django
- Android App Development – APSSDC

Languages

Telugu, English, Tamil (Basic)